

Python Programming - 2301CS404

Lab - 3

01) WAP to check whether the given number is positive or negative.

```
In [2]: num = int(input("Enter Number : "))

if (num > 0) : print(f"{num} Is Positive.")
elif (num < 0) : print(f"{num} Is Negative")
else : print(f"{num} Is Zero")
```

```
Enter Number : -4
-4 Is Negative
```

02) WAP to check whether the given number is odd or even.

```
In [4]: num = int(input("Enter Number : "))

if (num % 2 == 0) : print(f"{num} Is Even.")
else : print(f"{num} Is Odd")
```

```
Enter Number : 4
4 Is Even.
```

03) WAP to find out largest number from given two numbers using simple if and ternary operator.

```
In [16]: num = int(input("Enter Number : "))
num2 = int(input("Enter Number2 : "))

# if (num < num2) : print(f"{num2} Is Grater Than {num} .")
# else : print(f"{num} Is Grater Than {num2} .")

#or

max = (num,num2) [num < num2]
print("max : " ,max)

# or

# max = num if (num > num2) else num2
# print("max : " ,max)
```

```
Enter Number : 4
Enter Number2 : 5
```

04) WAP to find out largest number from given three numbers.

```
In [ ]: num1 = int(input("Enter Number1 : "))
num2 = int(input("Enter Number2 : "))
num3 = int(input("Enter Number3 : "))

# max = ((num1,num3) [num1 < num3],(num2,num3) [num2 < num3]) [num1 < num2]

# or

max = ( (num3) if (num3 > num2) else (num2) ) if (num1 < num3) else ( (num1) if

# or

# max_val = max(num1 , num2 , num3)

print("Max : ",max)
```

```
Enter Number1 : 73
Enter Number2 : 48
Enter Number3 : 3
Max : 73
```

05) WAP to check whether the given year is leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year]

```
In [ ]: year = int(input("Enter Year"))

if(year % 400 == 0 or (year % 4 == 0) and not (year % 100 == 0)):
    print(year,"Is Leap Year")
else:
    print(year,"Is Not Leap Year")
```

```
Enter Year 2025
2025 Is Not Leap Year
```

06) WAP in python to display the name of the day according to the number given by the user.

```
In [ ]: day = int(input("Enter Day (1-7)"))

match (day % 7):
    case 1 : print("Monday")
    case 2 : print("Tuesday")
    case 3 : print("Wednesday")
    case 4 : print("Thursday")
    case 5 : print("Friday")
    case 6 : print("Saturday")
    case 7 : print("Sunday")
```

```
Enter Day (1-7) 9
```

Tuesday

07) WAP to implement simple calculator which performs (add,sub,mul,div) of two no. based on user input.

```
In [ ]: num = int(input("Enter Number : "))
op = input("Enter Oparater ( + , - , * , / ) : ")
num2 = int(input("Enter Number2 : "))

match op:
    case "+" : print("Sum : ", num + num2)
    case "-" : print("Tuesday", num - num2)
    case "*" : print("Wednesday", num * num2)
    case "/" : print("Thursday", num / num2)
    case _   : print("Enter Valid Input")
```

```
Enter Number : 83
Enter Oparater ( + , - , * , / ) : *
Enter Number2 : 2
Wednesday 166
```

08) WAP to read marks of five subjects. Calculate percentage and print class accordingly.

Fail below 35

Pass Class between 35 to 45

Second Class

between 45 to 60

First Class between 60 to 70

Distinction if more than 70

```
In [ ]: mark1 , mark2 , mark3 , mark4 , mark5 = input("Enter 5 Subject Marks : ").split()

mark1 = int(mark1)
mark2 = int(mark2)
mark3 = int(mark3)
mark4 = int(mark4)
mark5 = int(mark5)

percentage = (mark1 + mark2 + mark3 + mark4 + mark5) / 5

if (percentage > 70):
    print("Distinction Class")

elif (percentage > 60):
    print("First Class")

elif (percentage > 45):
    print("Second Class")

elif (percentage > 35):
```

```
print("Pass Class")

else :
    print("Fail")
```

Enter 5 Subject Marks : 44 66 88 99 33
First Class

09) WAP to find the second largest number among three user input numbers.

```
In [ ]: num1 = int(input("Enter Number1 : "))
num2 = int(input("Enter Number2 : "))
num3 = int(input("Enter Number3 : "))

if (num1 >= num2 and num1 <= num3) or (num1 <= num2 and num1 >= num3):
    second = num1

elif (num2 >= num1 and num2 <= num3) or (num2 <= num1 and num2 >= num3):
    second = num2

else:
    second = num3

print("Second Largest Number Is:", second)
```

Enter Number1 : 727
Enter Number2 : 3
Enter Number3 : 4
Second Largest Number Is: 4

10) WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

- a. First 1 to 50 units – Rs. 2.60/unit
- b. Next 50 to 100 units – Rs. 3.25/unit
- c. Next 100 to 200 units – Rs. 5.26/unit
- d. above 200 units – Rs. 8.45/unit

```
In [ ]: units = int(input("Enter the number of units consumed: "))

bill = 0.0

if units <= 50:
    bill = units * 2.60

elif units <= 100:
    bill = (50 * 2.60) + (units - 50) * 3.25

elif units <= 200:
    bill = (50 * 2.60) + (50 * 3.25) + (units - 100) * 5.26

else:
    bill = (50 * 2.60) + (50 * 3.25) + (100 * 5.26) + (units - 200) * 8.45
```

```
print(f"Total Electricity Bill = Rs. {bill:.2f}")
```

Enter the number of units consumed: 1050

Total Electricity Bill = Rs. 8001.00